Probability

Probability defines the likelihood of occurrence of an event. There are many real-life situations in which we may have to predict the outcome of an event. We may be sure or not sure of the results of an event. In such cases, we say that there is a probability of this event to occur or not occur. Probability generally has great applications in games, in business to make predictions, and also it has extensive applications in this new area of artificial intelligence.

The probability of an event can be calculated by the probability formula by simply dividing the favourable number of outcomes by the total number of possible outcomes. The value of the probability of an event happening can lie between 0 and 1 because the favourable number of outcomes can never be more than the total number of outcomes. Also, the favorable number of outcomes cannot be negative. Let us discuss the basics of probability in detail in the following sections.

What is Probability?

Probability can be defined as the ratio of the number of favorable outcomes to the total number of outcomes of an event. For an experiment having 'n' number of outcomes, the number of favorable outcomes can be denoted by x. The formula to calculate the probability of an event is as follows.

Probability(Event) = Favorable Outcomes/Total Outcomes = x/n

Probability is used to predict the outcomes for the tossing of coins, rolling of dice, or drawing a card from a pack of playing cards. The probability is classified into two types:

- Theoretical probability
- Experimental probability

To understand each of these types, click on the respective links.

Terminology of Probability Theory

The following terms in probability theory help in a better understanding of the concepts of probability.

Experiment: A trial or an operation conducted to produce an outcome is called an experiment.

Sample Space: All the possible outcomes of an experiment together constitute a sample space. For example, the sample space of tossing a coin is {head, tail}.

Favorable Outcome: An event that has produced the desired result or expected event is called a favorable outcome. For example, when we roll two dice, the possible/favorable outcomes of getting the sum of numbers on the two dice as 4 are (1,3), (2,2), and (3,1).

Trial: A trial denotes doing a random experiment.

Random Experiment: An experiment that has a well-defined set of outcomes is called a random experiment. For example, when we toss a coin, we know that we would get ahead or tail, but we are not sure which one will appear.

Event: The total number of outcomes of a random experiment is called an event.

Equally Likely Events: Events that have the same chances or probability of occurring are called equally likely events. The outcome of one event is independent of the other. For example, when we toss a coin, there are equal chances of getting a head or a tail.

Exhaustive Events: When the set of all outcomes of an event is equal to the sample space, we call it an exhaustive event.

Mutually Exclusive Events: Events that cannot happen simultaneously are called mutually exclusive events. For example, the climate can be either hot or cold. We cannot experience the same weather simultaneously.

Events in Probability

In probability theory, an event is a set of outcomes of an experiment or a subset of the sample space. If P(E) represents the probability of an event E, then, we have,

- P(E) = 0 if and only if E is an impossible event.
- P(E) = 1 if and only if E is a certain event.
- 0 ≤ P(E) ≤ 1.

Suppose, we are given two events, "A" and "B", then the probability of event A, P(A) > P(B) if and only if event "A" is more likely to occur than the event "B". Sample space(S) is the set of all of the possible outcomes of an experiment and n(S) represents the number of outcomes in the sample space.

P(E) = n(E)/n(S)

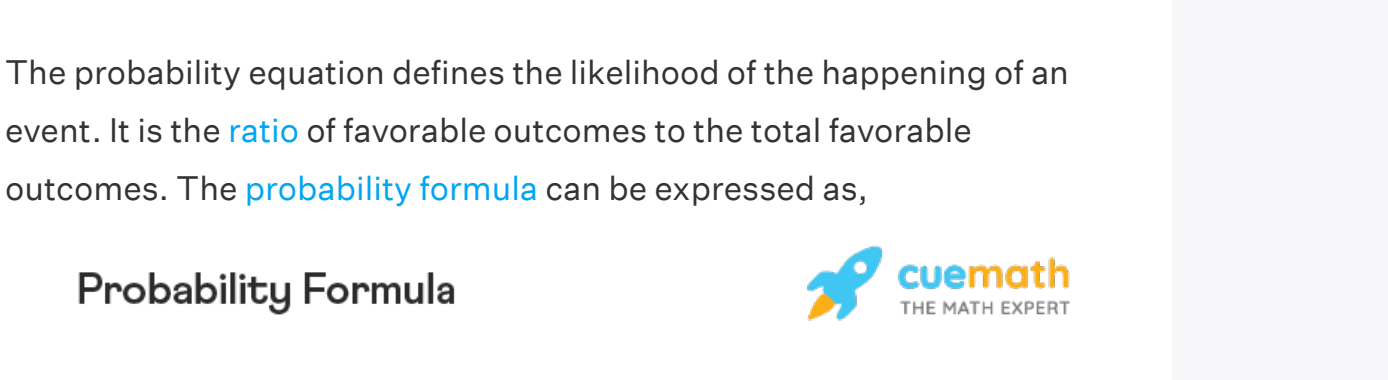
P(E') = (n(S) - n(E))/n(S) = 1 - (n(E)/n(S))

E' represents that the event will not occur.

Therefore, now we can also conclude that, P(E) + P(E') = 1

Probability Formula

The probability equation defines the likelihood of the happening of an event. It is the ratio of favorable outcomes to the total favorable outcomes. The probability formula can be expressed as,



$$P(A) = \frac{\text{Number of favorable outcomes to A}}{\text{Total number of possible outcomes}}$$

i.e., P(A) = n(A)/n(S)

where,

- P(A) is the probability of an event 'B'.
- n(A) is the number of favorable outcomes of an event 'B'.
- n(S) is the total number of events occurring in a sample space.

Different Probability Formulas

Probability formula with addition rule: Whenever an event is the union of two other events, say A and B, then
P(A or B) = P(A) + P(B) - P(A∩B)
P(A ∪ B) = P(A) + P(B) - P(A∩B)

Probability formula with the complementary rule: Whenever an event is the complement of another event, specifically, if A is an event, then P(not A) = 1 - P(A) or P(A') = 1 - P(A).
P(A) + P(A') = 1.

Probability formula with the conditional rule: When event A is already known to have occurred, the probability of event B is known as conditional probability and is given by:
P(B∣A) = P(A∩B)/P(A)

Probability formula with multiplication rule: Whenever an event is the intersection of two other events, that is, events A and B need to occur simultaneously. Then

- P(A ∩ B) = P(A)⋅P(B) (in case of independent events)
- P(A∩B) = P(A)⋅P(B∣A) (in case of dependent events)

Calculating Probability

In an experiment, the probability of an event is the possibility of that event occurring. The probability of any event is a value between (and including) "0" and "1". Follow the steps below for calculating probability of an event A:

- **Step 1:** Find the sample space of the experiment and count the elements. Denote it by n(S).
- **Step 2:** Find the number of favorable outcomes and denote it by n(A).
- **Step 3:** To find probability, divide n(A) by n(S). i.e., P(A) = n(A)/n(S).

Here are some examples that well describe the process of finding probability.

Example 1: Find the probability of getting a number less than 5 when a dice is rolled by using the probability formula.

Solution

To find:
Probability of getting a number less than 5
Given: Sample space, S = {1,2,3,4,5,6}
Therefore, n(S) = 6

Let A be the event of getting a number less than 5. Then A = {1,2,3,4}
So, n(A) = 4

Using the probability equation,
P(A) = (n(A))/(n(s))
p(A) = 4/6
m = 2/3

Answer: The probability of getting a number less than 5 is 2/3.

Example 2: What is the probability of getting a sum of 9 when two dice are thrown?

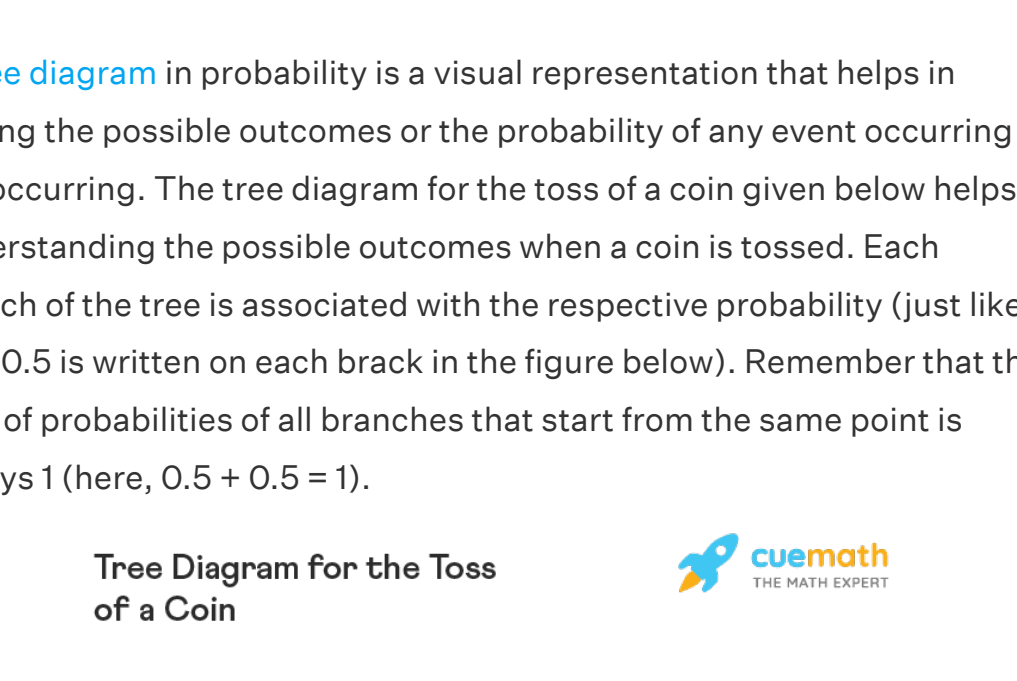
Solution:

There is a total of 36 possibilities when we throw two dice.
To get the desired outcome i.e., 9, we can have the following favorable outcomes.
(4,5),(5,4),(6,3)(3,6). There are 4 favorable outcomes.
Probability of an event P(E) = (Number of favorable outcomes) ÷ (Total outcomes in a sample space)
Probability of getting number 9 = 4 ÷ 36 = 1/9

Answer: Therefore the probability of getting a sum of 9 is 1/9.

Probability Tree Diagram

A tree diagram in probability is a visual representation that helps in finding the possible outcomes or the probability of any event occurring or not occurring. The tree diagram for the toss of a coin given below helps in understanding the possible outcomes when a coin is tossed. Each branch of the tree is associated with the respective probability (just like how 0.5 is written on each brack in the figure below). Remember that the sum of probabilities of all branches that start from the same point is always 1 (here, 0.5 + 0.5 = 1).



Types of Probability

There can be different perspectives or types of probabilities based on the nature of the outcome or the approach followed while finding probability of an event happening. The four types of probabilities are,

- Classical Probability
- Empirical Probability
- Subjective Probability
- Axiomatic Probability

Classical Probability

Classical probability, often referred to as the "priori" or "theoretical probability", states that in an experiment where there are B equally likely outcomes, and event X has exactly A of these outcomes, then the probability of X is A/B, or P(X) = A/B. For example, when a fair die is rolled, there are six possible outcomes that are equally likely. That means, there is a 1/6 probability of rolling each number on the die.

Empirical Probability

The empirical probability or the experimental perspective evaluates probability through thought experiments. For example, if a weighted die is rolled, such that we don't know which side has the weight, then we can get an idea for the probability of each outcome by rolling the die number of times and calculating the proportion of times the die gives that outcome and thus find the probability of that outcome.

Subjective Probability

Subjective probability considers an individual's own belief of an event occurring. For example, the probability of a particular team winning a football match on a fan's opinion is more dependent upon their own belief and feeling and not on a formal mathematical calculation.

Axiomatic Probability

In axiomatic probability, a set of rules or axioms by Kolmogorov are applied to all the types. The chances of occurrence or non-occurrence of any event can be quantified by the applications of these axioms, given as,

- The smallest possible probability is zero, and the largest is one.
- An event that is certain has a probability equal to one.
- Any two mutually exclusive events cannot occur simultaneously, while the union of events says only one of them can occur.

Coin Toss Probability

Let us now look into the probability of tossing a coin. Quite often in games like cricket, for making a decision as to who would bowl or bat first, we sometimes use the tossing of a coin and decide based on the outcome of the toss. Let us check how we can use the concept of probability in the tossing of a single coin. Further, we shall also look into the tossing of two and three coins.

Tossing a Coin

A single coin on tossing has two outcomes, a head, and a tail. The concept of probability which is the ratio of favorable outcomes to the total number of outcomes can be used in finding probability of getting the head and the probability of getting a tail.

Total number of possible outcomes = 2; Sample Space = {H, T}; H: Head, T: Tail

- P(H) = Number of heads/Total outcomes = 1/2
- P(T)= Number of Tails/ Total outcomes = 1/2

Tossing Two Coins

In the process of tossing two coins, we have a total of four (= 2^2) outcomes. The probability formula can be used to find the probability of two heads, one head, no head, and a similar probability can be calculated for the number of tails. The probability calculations for the two heads are as follows.

Total number of outcomes = 4; Sample Space = {(H, H), (H, T), (T, H), (T, T)}

- P(2H) = P(0 T) = Number of outcome with two heads/Total Outcomes = 1/4
- P(1H) = P(1T) = Number of outcomes with only one head/Total Outcomes = 2/4 = 1/2
- P(0H) = (2T) = Number of outcome with two heads/Total Outcomes = 1/4

Tossing Three Coins

The number of total outcomes on tossing three coins simultaneously is equal to 2^3 = 8. For these outcomes, we can find the probability of getting one head, two heads, three heads, and no head. A similar probability can also be calculated for the number of tails.

Total number of outcomes = 2^3 = 8 Sample Space = {(H, H, H), (H, H, T), (H, T, H), (T, H, H), (T, T, H), (T, H, T), (H, T, T), (T, T, T)}

- P(0H) = P(3T) = Number of outcomes with no heads/Total Outcomes = 1/8
- P(1H) = P(2T) = Number of Outcomes with one head/Total Outcomes = 3/8
- P(2H) = P(1T) = Number of outcomes with two heads /Total Outcomes = 3/8
- P(3H) = P(0T) = Number of outcomes with three heads/Total Outcomes = 1/8

Dice Roll Probability

Many games use dice to decide the moves of players across the games. A dice has six possible outcomes and the outcomes of a dice is a game of chance and can be obtained by using the concepts of probability. Some games also use two dice, and there are numerous probabilities that can be calculated for outcomes using two dice. Let us now check the outcomes, their probabilities for one dice and two dice respectively.

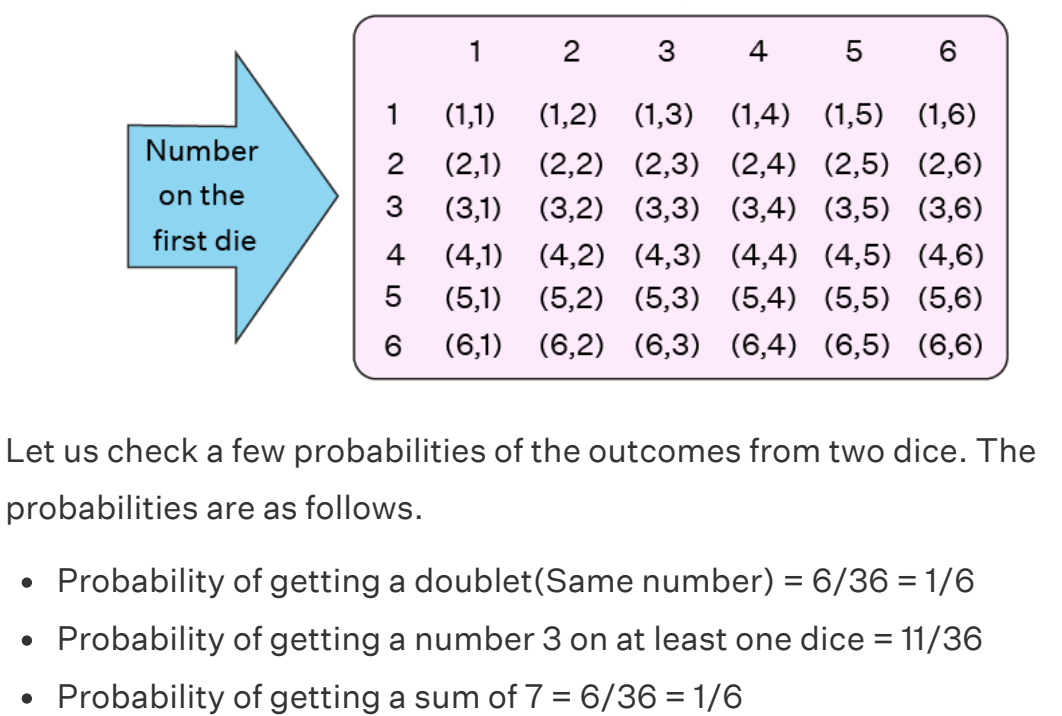
Rolling One Dice

The total number of outcomes on rolling a die is 6, and the sample space is {1, 2, 3, 4, 5, 6}. Here we shall compute the following few probabilities to help in better understanding the concept of probability on rolling one dice.

- P(Even Number) = Number of even number outcomes/Total Outcomes = 3/6 = 1/2
- P(Odd Number) = Number of odd number outcomes/Total Outcomes = 3/6 = 1/2
- P(Prime Number) = Number of prime number outcomes/Total Outcomes = 3/6 = 1/2

Rolling Two Dice

The total number of outcomes on rolling two dice is 6^2 = 36. The following image shows the sample space of 36 outcomes on rolling two dice.


Sample Space for Tossing Two Coins
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Number on the second die
Number on the first die

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

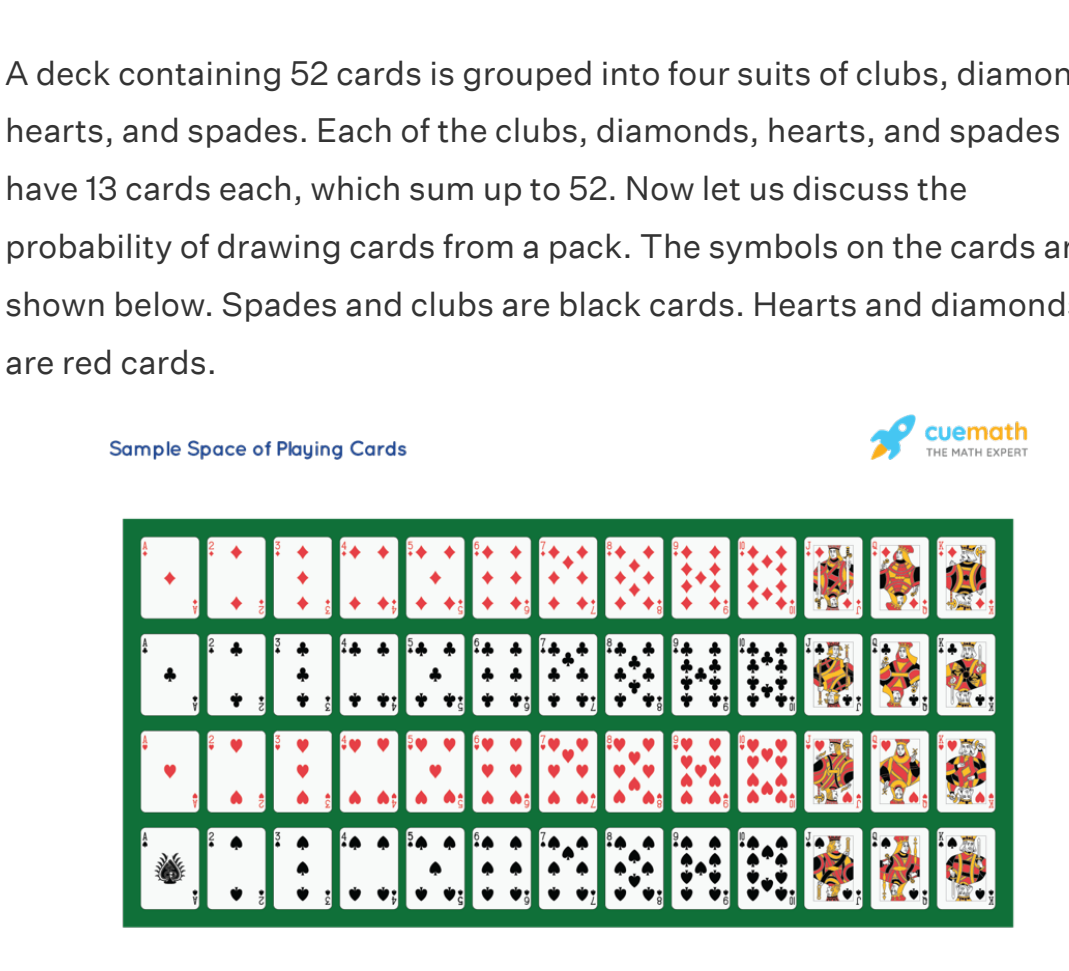
Let us check a few probabilities of the outcomes from two dice. The probabilities are as follows.

- Probability of getting a doublet(Same number) = 6/36 = 1/6
- Probability of getting a number 3 on at least one dice = 11/36
- Probability of getting a sum of 7 = 6/36 = 1/6

As we see, when we roll a single die, there are 6 possibilities. When we roll two dice, there are 36 (= 6^2) possibilities. When we roll 3 dice we get 216 (= 6^3) possibilities. So a general formula to represent the number of outcomes on rolling 'n' dice is 6^n.

Probability of Drawing Cards

A deck containing 52 cards is grouped into four suits of clubs, diamonds, hearts, and spades. Each of the clubs, diamonds, hearts, and spades have 13 cards each, which sum up to 52. Now let us discuss the probability of drawing cards from a pack. The symbols on the cards are shown below. Spades and clubs are black cards. Hearts and diamonds are red cards.


Sample Space of Playing Cards
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The 13 cards in each suit are ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, jack, queen, king. In these, the jack, the queen, and the king are called face cards. We can understand the card probability from the following examples.

- The probability of drawing a black card is P(Black card) = 26/52 = 1/2
- The probability of drawing a hearts card is P(Hearts) = 13/52 = 1/4
- The probability of drawing a face card is P(Face card) = 12/52 = 3/13
- The probability of drawing a card numbered 4 is P(4) = 4/52 = 1/13
- The probability of drawing a red card numbered 4 is P(4 Red) = 2/52 = 1/26

Probability Theorems

The following theorems of probability are helpful to understand the applications of probability and also perform the numerous calculations involving probability.

Theorem 1: The sum of the probability of happening of an event and not happening of an event is equal to 1. P(A) + P(A') = 1.

Theorem 2: The probability of an impossible event or the probability of an event not happening is always equal to 0. P(ϕ) = 0.

Theorem 3: The probability of a sure event is always equal to 1. P(A) = 1

Theorem 4: The probability of happening of any event always lies between 0 and 1. 0 < P(A) < 1

Theorem 5: If there are two events A and B, we can apply the formula of the union of two sets and we can derive the formula for the probability of happening of event A or event B as follows.

P(A∪B) = P(A) + P(B) - P(A∩B)

Also for two mutually exclusive events A and B, we have P(A U B) = P(A) + P(B)

Bayes' Theorem on Conditional Probability

Bayes' theorem describes the probability of an event based on the condition of occurrence of other events. It is also called conditional probability. It helps in calculating the probability of happening of one event based on the condition of happening of another event.

For example, let us assume that there are three bags with each bag containing some blue, green, and yellow balls. What is the probability of picking a yellow ball from the third bag? Since there are blue and green colored balls also, we can arrive at the probability based on these conditions also. Such a probability is called conditional probability.

The formula for Bayes' theorem is $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

where, $P(A|B)$ denotes how often event A happens on a condition that B happens.

where, $P(B|A)$ denotes how often event B happens on a condition that A happens.

$P(A)$ the likelihood of occurrence of event A.

$P(B)$ the likelihood of occurrence of event B.

Law of Total Probability

If there are n number of events in an experiment, then the sum of the probabilities of those n events is always equal to 1.

$P(A_1) + P(A_2) + P(A_3) + … + P(A_n) = 1$

Important Notes on Probability:

- Probability is a measure of how likely an event is to happen.
- Probability is represented as a fraction and always lies between 0 and 1.
- An event can be defined as a subset of sample space.
- The sample of throwing a coin is {head, tail} and the sample space of throwing dice is {1, 2, 3, 4, 5, 6}.
- A random experiment cannot predict the exact outcomes but only some probable outcomes.

Probability Examples

Example 1: What is the probability of getting a sum of 10 when two dice are thrown?

Solution:

There are 36 possibilities when we throw two dice.

The desired outcome is 10. To get 10, we can have three favorable outcomes.

{(4,6),(6,4),(5,5)}

Probability of an event = number of favorable outcomes/ sample space

Probability of getting number 10 = 3/36 =1/12

Answer: Therefore the probability of getting a sum of 10 is 1/12.

Example 2: In a bag, there are 6 blue balls and 8 yellow balls. One ball is selected randomly from the bag. Find the probability of getting a blue ball.

Solution:

Let us assume the probability of drawing a blue ball to be P(B)

Number of favorable outcomes to get a blue ball = 6

Total number of balls in the bag = 14

P(B) = Number of favorable outcomes/Total number of outcomes = 6/14 = 3/7

Answer: Therefore the probability of drawing a blue ball is 3/7.

Example 3: There are 5 cards numbered: 2, 3, 4, 5, 6. Find the probability of picking a prime number, and putting it back, you pick a composite number.

Solution:

The two events are independent. Thus we use the product of the probability of the events.

P(getting a prime) = n(favorable events)/ n(sample space) = {2, 3, 5}/{2, 3, 4, 5, 6} = 3/5

p(getting a composite) = n(favorable events)/ n(sample space) = {4, 6}/{2, 3, 4, 5, 6}= 2/5

Thus the total probability of the two independent events = P(prime) × P(composite)

= 3/5 × (2/5)

= 6/25

Answer: Therefore the probability of picking a prime number and a prime number again is 6/25.

Example 4: Find the probability of getting a face card from a standard deck of cards using the probability equation.

Solution: To find:
Probability of getting a face card
Given: Total number of cards = 52
Number of face cards = Favorable outcomes = 12
Using the probability formula,
Probability = (Favorable Outcomes)÷(Total Favorable Outcomes)
P(face card) = 12/52
m = 3/13

Answer: The probability of getting a face card is 3/13

FAQs on Probability

What is the Meaning of Probability in Statistics?

Probability is a branch of math which deals with finding out the likelihood of the occurrence of an event. Probability measures the chance of an event happening and is equal to the number of favorable events divided by the total number of events. The value of probability ranges between 0 and 1, where 0 denotes uncertainty and 1 denotes certainty.

How to Find Probability?

The probability can be found by first knowing the sample space of the outcomes of an experiment. A probability is generally calculated for an event (x) within the sample space. The probability of an event happening is obtained by dividing the number of outcomes of an event by the total number of possible outcomes or sample space.

What are the Three Types of Probability?

The three types of probabilities are theoretical probability, experimental probability, and axiomatic probability. The theoretical probability calculates the probability based on formulas and input values. The experimental probability gives a realistic value and is based on the experimental values for calculation. Quite often the theoretical and experimental probability differ in their results. And the axiomatic probability is based on the axioms which govern the concepts of probability.

How To Calculate Probability?

The probability of any event depends upon the number of favorable outcomes and the total outcomes. Finding probability is finding the ratio of the number of favorable outcomes to the total outcomes in that sample space. It is expressed as, Probability of an event P(E) = (Number of favorable outcomes) ÷ (Number of Elements in Sample space).

What is Conditional Probability?

The conditional probability predicts the happening of one event based on the happening of another event. If there are two events A and B, conditional probability is a chance of occurrence of event B provided the event A has already occurred. The formula for the conditional probability of happening of event B, given that event A, has happened is $P(B/A) = P(A \cap B)/P(A)$.

What is Experimental Probability?

The experimental probability is based on the results and the values obtained from the probability experiments. Experimental probability is defined as the ratio of the total number of times an event has occurred to the total number of trials conducted. The results of the experimental probability are based on real-life instances and may differ in values from theoretical probability.

What is a Probability Distribution?

The two important probability distributions are binomial distribution and Poisson distribution. The binomial distribution is defined for events with two probability outcomes and for events with a multiple number of times of such events. The Poisson distribution is based on the numerous probability outcomes in a limited space of time, distance, sample space. An example of the binomial distribution is the tossing of a coin with two outcomes, and for conducting such a tossing experiment with n number of coins. A Poisson distribution is for events such as antigen detection in a plasma sample, where the probabilities are numerous.

How are Probability and Statistics Related?

The probability calculates the happening of an experiment and it calculates the happening of a particular event with respect to the entire set of events. For simple events of a few numbers of events, it is easy to calculate the probability. But for calculating probabilities involving numerous events and to manage huge data relating to those events we need the help of statistics. Statistics helps in rightly analyzing

How Probability is Used in Real Life?

Probability has huge applications in games and analysis. Also in real life and industry areas where it is about prediction we make use of probability. The prediction of the price of a stock, or the performance of a team in cricket requires the use of probability concepts. Further, the new technology field of artificial intelligence is extensively based on probability.

Where Do We Use the Probability Formula In Our Real Life?

The following activities in our real-life tend to follow the probability equation:

- Weather forecasting
- Playing cards
- Voting strategy in politics
- Rolling a dice.
- Pulling out the exact matching socks of the same color
- Chances of winning or losing in any sports.

How was Probability Discovered?

The use of the word "probable" started first in the seventeenth century when it was referred to actions or opinions which were held by sensible people. Further, the word probable in the legal content was referred to a proposition that had tangible proof. The field of permutations and combinations, statistical inference, cryptoanalysis, frequency analysis have altogether contributed to this current field of probability.

What is the Conditional Probability Formula?

The conditional probability depends upon the happening of one event based on the happening of another event. The conditional probability formula of happening of event B, given that event A, has already happened is expressed as $P(B/A) = P(A \cap B)/P(A)$.